

Excluding the Bounded Negative Part

Lean-checked rigidity for Erdős Problem #243

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ABSTRACT

Sylvester's sequence satisfies $a_{n+1} = a_n^2 - a_n + 1$ and has reciprocal sum 1. Erdős asked whether every increasing integer sequence with $a_{n+1}/a_n^2 \rightarrow 1$ and rational reciprocal sum satisfies this recurrence eventually. Write E_n for the centred denominator-clearing error in Koizumi's integer state coordinates. Our main result excludes the entire bounded-negative regime: under the exact C/D dynamics, positivity, normalised vanishing, and an eventually bounded negative part, $E_n = 0$ eventually, so the Sylvester recurrence holds eventually. Normalised vanishing itself yields the required strict centring after a finite shift.

The proof has three arithmetic steps. Bounded negative errors give bounded upward increments of C_n . On a hypothetical tail with no zero error, integrality and normalised vanishing force $C_n \rightarrow \infty$; the tail gcd then stabilises. After reduction, the successive multipliers are pairwise coprime and each is coprime to all later numerators; a block of consecutive multiples contradicts these coprimality conditions. Finally, $E_n = 0$ is absorbing. The same argument in summable form gives a second criterion: under the exact dynamics, positivity, the strict centred-step equation, normalised vanishing, and $\sum_n (-E_n)_+/C_n < \infty$, the Sylvester recurrence is recovered [the finite-negative-mass recovery theorem](#). The summability hypothesis is not known for the canonical orbit. An ordinary first-crossing argument in Section 10 gives smaller equivalent sufficient sums, restricted to LCM numerator records and allowing logarithmic discounts; their finiteness remains open as well.

Several sharper exclusions explain why the remaining case is genuinely unbounded. A negative error cannot be constant, at any magnitude or scale, and cannot be periodic when $|E_n| < a_n$ and the drift is positive. If $E_n \geq 0$, the state is nonincreasing and therefore eventually zero [nonnegative-error descent](#); the recurrence then follows from [eventual Sylvester recurrence](#). On a reduced recovery interval of length L , deleted old moduli satisfy the exact inequality

$$K^L \operatorname{lcm}(m_q : q \in R)^2 < (K + 1)^L.$$

Hence fixed-length recoveries are eventually cancellation-free, although recoveries of increasing length remain possible.

Koizumi's pseudo-greedy coordinates supply normalised vanishing and a stronger centring range for the canonical Erdős orbit, but neither boundedness nor summability of its negative part. Thus any surviving orbit must have cofinally unbounded negative excursions and divergent normalised negative mass. Excluding that case is the exact open boundary, and Problem #243 remains open.

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

1 Introduction

Sylvester's sequence 2, 3, 7, 43, 1807, ... is generated by $a_n = a_{n-1}^2 - a_{n-1} + 1$ and has reciprocal sum 1; the recurrence is exactly the statement that the tail beyond a_{n-1} equals $1/(a_{n-1} - 1)$. Erdős Problem #243 asserts that this is the only way a sequence of that growth can have a rational reciprocal sum:

Problem 1.1 (Erdős #243). Let $1 \leq a_1 < a_2 < \dots$ be a sequence of integers with

$$\lim_{n \rightarrow \infty} \frac{a_n}{a_{n-1}^2} = 1 \quad \text{and} \quad \sum \frac{1}{a_n} \in \mathbb{Q}.$$

Then $a_n = a_{n-1}^2 - a_{n-1} + 1$ for all sufficiently large n .

See [4, p. 64] and [5, p. 105]. Bloom's catalogue records the problem as open [8]. Below we index the same recurrence as $a_{n+1} = a_n^2 - a_n + 1$, which is the shift the formal sources use; the two forms are the same statement. Write $\text{syl}(a) = a^2 - a + 1$, the [Sylvester successor](#).

Relation to prior work. Two published results reach the original problem under analytic hypotheses that are not assumed below. Their hypotheses are conditions on the sequence (a_n) itself. Duverney's is not comparable with the state-level hypotheses used here; the Erdős–Straus one, read through the dictionary of the next paragraph, is implied by nonnegativity of E , the hypothesis of Theorem 4.1, so that criterion already covers the descent proved below (see [7, Cor. 20(1) and Prop. 19(1), pp. 12–13]). Erdős and Straus proved that if $\lim a_n/a_{n-1}^2 = 1$, the reciprocal sum is rational, and a_n does not satisfy the recurrence, then

$$\limsup_{n \rightarrow \infty} \frac{[a_1, \dots, a_n]}{a_{n+1}} \left(\frac{a_{n+1}^2}{a_{n+2}} - 1 \right) > 0,$$

where $[a_1, \dots, a_n]$ is the least common multiple [1, Theorem 3, p. 132]. This is the indexing in the original theorem. Erdős's 1988 survey, followed by the current catalogue summary, instead prints $a_n^2/a_{n+1} - 1$ in the second factor while retaining the same prefix-LCM quotient; that displayed summary is off by one and is not followed here [5, p. 105][8]. Koizumi records the convenient sufficient rate $a_n^2/a_{n+1} = 1 + o(1/n)$ under which the Erdős–Straus criterion settles the problem [7, Remark 21, p. 13].

Duverney proved a conditional signed form of the problem itself: for signs $\epsilon_n \in \{-1, 1\}$, if $\sum_{n \geq 0} (a_{n+1}/a_n^2 - 1)$ converges, then the reciprocal sum with numerators ϵ_n is rational if and only if

$$a_{n+1} = a_n^2 - (\epsilon_{n+1}/\epsilon_n)a_n + \epsilon_{n+2}/\epsilon_{n+1}$$

for all large n [2, Corollary 3.2, p. 287]. Absolute convergence is a stronger sufficient specialisation, not the hypothesis printed in that corollary. The all-positive specialisation is the form relevant here.

Badea's positive-term criterion is adjacent but different: for positive rational terms b_n/a_n , eventual strict inequality

$$a_{n+1} > \frac{b_{n+1}}{b_n} a_n^2 - \frac{b_{n+1}}{b_n} a_n + 1$$

forces irrationality, while rationality under the corresponding non-strict inequality forces eventual equality [3, Theorem A, p. 313; Cor. 2.2, p. 316]. These positive-term hypotheses do not cover the mixed-sign state orbits studied here. Neither result is formalised here, and the results below do not recover either of them.

Koizumi's work on doubly exponential sequences [7] bears directly on the state system used below, and supplies the coordinates in which several of the statements of this note are most naturally read. For a rational $r = p/q$ with pseudo-greedy expansion (a_n) , remainders (x_n) and gap sequence $\varepsilon_n = x_n^{-1} + 1 - a_n$, his Lemma 15 [7, p. 9] produces integers $c_n > 0$, d_n and e_n with $x_n = c_n/d_n$ and $\varepsilon_n = e_n/c_n$, satisfying

$$a_n = \frac{d_n - e_n}{c_n} + 1, \quad c_{n+1} = c_n - e_n, \quad d_{n+1} = a_n d_n,$$

and the proof of that lemma also gives $c_{n+1} = a_n c_n - d_n$. Under the dictionary $(C_n, D_n, E_n) = (c_n, d_n, e_n)$ these are exactly the objects of Section 2: the first relation is the centring map ctr rewritten as $e_n = d_n - (a_n - 1)c_n$, the second is Proposition 2.1, the third is the denominator update, and the fourth is the tail update. The lower-case e_n of Section 2 is $-E_n$, so it is the negative of Koizumi's e_n at an index where the centred state is negative.

His Theorem 16 [7, p. 11] shows that his Conjecture 6 — that for a positive rational r whose gap sequence satisfies $\varepsilon_n \rightarrow 0$ one has $\varepsilon_n = 0$ for all large n — is equivalent to an affirmative answer to the question of Erdős and Graham recorded as his Question 5, which is Problem 1.1 above. Two of the implications proved below have prior art in these canonical coordinates: his Lemma 13, that $\varepsilon_n = 0$ forces $\varepsilon_{n+1} = 0$, is the absorption of Theorem 3.7, and his Proposition 19(2), that $\varepsilon_n \geq 0$ for all large n forces $\varepsilon_n = 0$ for all large n , is the descent of Theorem 4.1. These appear in canonical coordinates in [7, Lemma 13, p. 9; Prop. 19(2), p. 12].

The growth hypothesis in Problem 1.1 is calibrated by two facts about Sylvester's sequence. It satisfies $a_n \approx c_0^{2^n}$ with $c_0 = 1.2640847\dots$, and shifting it further produces sequences with $a_n \approx C^{2^n}$ for arbitrarily large C whose reciprocals still sum to a rational number [6, arXiv v4, p. 2]. The classical sufficient condition for irrationality, $\lim_n a_n^{1/2^n} = \infty$, is therefore sharp. Kovač and Tao identify that condition as folklore [6, arXiv v4, p. 2]; they attribute the sharpness observation to Erdős (1975). A sequence with $a_n/a_{n-1}^2 \rightarrow 1$ has $a_n^{1/2^n}$ convergent, so that criterion says nothing about the sequences of Problem 1.1, and rationality is genuinely possible there. Sylvester's sequence is A000058 in the OEIS and its shifts A129871. For the recent literature on irrationality of Ahmes series we refer to Kovač and Tao [6], who resolve several problems of Erdős and Graham drawn from the same two sources cited above, and whose introduction gives a sample of the intermediate work, including Sándor (1984) and Badea (1987). They do not treat Problem #243; the rigidity conclusion asked for there is not among their results.

The *Formal Conjectures* collection contains a mathematically equivalent unproved declaration, up to its zero-based indexing [9]. Its summand is $\mathbb{Q}$ -valued, so Lean's Summable hypothesis asserts the existence of a sum in $\mathbb{Q}$; the finite indexing shift changes that sum only by a rational prefix. The declaration therefore does encode the rationality premise, but its proof is sorry. Its role here is statement-level prior art; it supplies no proof authority, and the development below is independent of it. The machine-checked results concern the state system: they exclude constant and periodic negative magnitudes, the bounded-rise barrier of Theorem 7.3, and the two conditional endpoints (Theorems 8.1 and 9.1), each stated with the exact hypotheses that separate it from Problem #243.

The hypothesis is asymptotic and the conclusion is exact, so the argument must convert an analytic rate into an integer obstruction. The conversion used here is the classical one: clear denominators along the sequence, so that rationality makes a tail integral, and then centre that integer at the value it would take on Sylvester's sequence. What remains is a single integer error E , and the whole question is whether E can avoid vanishing.

Status. The original problem remains open. Statements marked as Lean-checked refer to the linked propositions accepted by the pinned kernel, with no sorry, added axioms, or unchecked evaluation. The note states the remaining mathematical obligations explicitly.

Companion system context. The [claim and trust boundary](#), [cold-clone route to proof authority](#), and [public contribution protocol](#) are described in sibling papers.

Strongest result. Under the exact integer dynamics, positivity, normalised vanishing, and an eventually bounded negative part, the centred error E_n is eventually zero. The original sequence therefore satisfies the Sylvester recurrence from some point onward (Theorem 8.1).

Second endpoint. The same conclusion holds when the normalised negative mass $\sum_n (-E_n)_+ / C_n$ is finite, provided the strict centred-step equation is available (Theorem 9.1).

What remains open. Koizumi's coordinates provide normalised vanishing for the canonical orbit, but they do not provide either bounded negative excursions or finite normalised negative mass. Any surviving counterexample must therefore have cofinally unbounded negative excursions and divergent normalised negative mass (Proposition 11.1).

Proof spine. Bounded negative errors control the rise of the integer tail state. A gcd stabilisation reduces the orbit to pairwise-coprime fresh moduli, and a block of consecutive multiples then gives the contradiction. The finite-mass argument replaces the uniform bound by a summable recovery estimate.

The remaining exclusions sharpen this boundary. By Theorem 3.7, a counterexample has $E_n \neq 0$ for every large n . By Theorem 4.1 it cannot have E eventually nonnegative. If its tail is all negative, Theorems 5.2 and 6.1 exclude, respectively, constant magnitude and periodic magnitude in the stated regime $e_n < a_n$ with positive drift; they are not a classification of a mixed-sign tail. Under normalised vanishing, Theorems 8.1 and 9.1 show that the negative part is neither bounded nor of finite normalised mass. What survives is stated in Section 11.

Structure

Section 2 sets up the state system, and Section 3 proves the defect identity, from which eventual vanishing of E forces the Sylvester recurrence and a single vanishing error propagates. Section 4 disposes of the case $E \geq 0$ in three lines. The rest of the note is the case $E < 0$, in four widening steps: constant magnitude (Section 5), periodic magnitude (Section 6), bounded magnitude via a coprimality barrier (Sections 7 and 8), finite normalised negative mass (Section 9), and what is left (Section 11). These sections contain the new content; a reader who wants only the strongest statements should read Sections 8 and 9, and then Section 11. Appendix A is a guide to the formal sources, and Appendix B records an exact residue reduction that an earlier finite search used. Linked phrases open the corresponding Lean declaration at the pinned source revision 99f4bf47422a.

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2 The centred integer state

The three maps defined next carry out the two steps announced at the end of Section 1. The first two clear denominators along the sequence, so that a rational reciprocal sum leaves a sequence of integers; the third subtracts from that integer the value it takes on Sylvester's sequence, leaving one number whose vanishing is the whole question. All three are maps between integer tuples and carry no hypothesis of any kind.

For $a, D, C \in \mathbb{Z}$ define

$$\text{den}(a, D) = aD, \quad \text{tail}(a, D, C) = aC - D, \quad \text{ctr}(a, D, C) = D - (a - 1)C,$$

the [denominator update](#), the [tail update](#), and the [centred state](#). Given a sequence (a_n) of integers, whose terms we call the *multipliers*, we write $D_{n+1} = \text{den}(a_n, D_n)$ for the *denominator state*, $C_{n+1} = \text{tail}(a_n, D_n, C_n)$ for the *tail state*, and $E_n = \text{ctr}(a_n, D_n, C_n)$ for the *centred state*, also called the *error*. The word *centred* records what the third map does: it measures D_n against $(a_n - 1)C_n$, and $D_n = (a_n - 1)C_n$ holds at every index on Sylvester's sequence (Example 2.2), so E is a deviation from that sequence rather than a size. At an index where $E_n < 0$ we write $e_n = -E_n > 0$ and call it the *magnitude* of the error there.

Proposition 2.1 (update law). $\text{tail}(a, D, C) = C - \text{ctr}(a, D, C)$; that is, $C_{n+1} = C_n - E_n$.

Proof. $C - (D - (a - 1)C) = aC - D$. □

Formalised as the [update law](#). The state therefore moves by exactly the error at each step, which is what makes the sign of E decisive: where E is positive the state falls, where E is negative it rises, and where E vanishes it is stationary.

Example 2.2 (the Sylvester orbit). Take $a_n = 2, 3, 7, 43, 1807, \dots$ with $a_{n+1} = a_n^2 - a_n + 1$, and start the state at $D_0 = C_0 = 1$. The two updates give

n	a_n	D_n	C_n	E_n
0	2	1	1	0
1	3	2	1	0
2	7	6	1	0
3	43	42	1	0
4	1807	1806	1	0

and $D_n = a_n - 1$ with $C_n = 1$ at every index: if $D_n = a_n - 1$ and $C_n = 1$ then $C_{n+1} = a_n - (a_n - 1) = 1$ and $D_{n+1} = a_n(a_n - 1) = a_{n+1} - 1$. Hence $E_n = D_n - (a_n - 1)C_n = 0$ throughout and the tail state never moves, which is Proposition 2.1 in the stationary case.

The intended reading, which is not formalised. Let $T_n = \sum_{k \geq n} 1/a_k$ and suppose $T_0 \in \mathbb{Q}$. Put D_0 equal to a common denominator and $D_n = D_0 a_0 \cdots a_{n-1}$, and set $C_n = D_n T_n$. Then $C_n \in \mathbb{Z}$ for every n , and from $T_n = 1/a_n + T_{n+1}$ one gets $C_{n+1} = a_n C_n - D_n$, which is the tail update. On Sylvester's sequence $T_n = 1/(a_n - 1)$ exactly, so $D_n = (a_n - 1)C_n$ and $E_n = 0$: the centred state measures deviation from the Sylvester tail identity, and it vanishes identically on the Sylvester orbit.

None of this paragraph is formalised. The Lean module contains no series, no rationality hypothesis, and no growth hypothesis; it proves identities and implications about the integer system above, and about its natural-number realisation, for which the bridging identities are the [tail realisation](#), the [denominator realisation](#), and the [signed update law](#). The identification with reciprocal tails motivates the definitions. The formal development does not check that identification.

Proposition 2.3 (scale equivariance). *For every $s, a, D, C \in \mathbb{Z}$,*

$$\text{den}(a, sD) = s \text{ den}(a, D), \quad \text{tail}(a, sD, sC) = s \text{ tail}(a, D, C),$$

$$\text{ctr}(a, sD, sC) = s \text{ ctr}(a, D, C).$$

Formalised as the [denominator equivariance](#), the [tail equivariance](#), and the [centred equivariance](#). The system depends only on the ray through (D, C) . This is used twice below: once to normalise the finite search of Appendix B, and once, in Theorem 6.1, as the induction step that removes common scale.

3 The defect identity and its two consequences

Section 2 connects the Sylvester recurrence to the error in one direction only: on Sylvester's sequence the error vanishes. The converse is what the problem needs, and it comes from a single identity, which expresses the deviation of a_{n+1} from the Sylvester successor, multiplied by the tail state C_{n+1} , as a combination of the errors at n and at $n+1$, the first weighted by a_n^2 . Its two consequences are Theorem 3.6, in which an eventually vanishing error forces the recurrence provided the tail state is eventually nonzero, and

Theorem 3.7, in which, under strict centring (the condition $|E_n| < C_n$ at every index), one vanishing error propagates to every later index.

Write $\text{def}(a, a') = a' - \text{syl}(a)$ for the deviation of the next term from the Sylvester successor, the [Sylvester defect](#).

Theorem 3.1 (defect identity). *For all $a, a', D, C \in \mathbb{Z}$,*

$$\text{def}(a, a') \cdot \text{tail}(a, D, C) = a^2 \text{ctr}(a, D, C) - \text{ctr}(a', \text{den}(a, D), \text{tail}(a, D, C)),$$

that is, $\Delta_n C_{n+1} = a_n^2 E_n - E_{n+1}$.

Proof. A direct expansion: both sides equal $a'aC - a'D - a^3C + a^2D + a^2C - aD - aC + D$. $\square$

Formalised as the [defect identity](#). It is a polynomial identity in four indeterminates, with no hypothesis of any kind, and it carries the whole argument twice over: once for rigidity, and once for absorption.

Example 3.2 (one defect and the error it creates). Continue Example 2.2 but replace $a_3 = 43$ by $a_3 = 44$. The states $D_3 = 42$ and $C_3 = 1$ are unchanged, since they depend only on a_0, a_1, a_2 , and $E_2 = 0$ still. The defect is $\Delta_2 = a_3 - \text{syl}(a_2) = 44 - 43 = 1$, and $E_3 = D_3 - (a_3 - 1)C_3 = 42 - 43 = -1$, so the identity reads

$$\Delta_2 C_3 = 1 \cdot 1 = 1 = 49 \cdot 0 - (-1) = a_2^2 E_2 - E_3.$$

By Proposition 2.1 the tail state then rises, $C_4 = C_3 - E_3 = 2$. A single unit of defect at one index has produced a negative error of magnitude 1 at the next, and the state has started to climb.

Theorem 3.3 (two vanishing errors force the step). *Let $a, a', D, C \in \mathbb{Z}$ with $\text{tail}(a, D, C) \neq 0$. If $\text{ctr}(a, D, C) = 0$ and $\text{ctr}(a', \text{den}(a, D), \text{tail}(a, D, C)) = 0$, then $a' = \text{syl}(a)$.*

Proof. By Theorem 3.1 the product $\text{def}(a, a') \cdot \text{tail}(a, D, C)$ is zero, and the second factor is nonzero by hypothesis. $\square$

Formalised as the [local rigidity step](#). The hypothesis $C_{n+1} \neq 0$ is not removable: at $C_{n+1} = 0$ the identity gives no information about a' .

3.1 Two-step cancellation and feedback

The one-step state equations hide a second-order constraint once the raw numerator is reduced at two consecutive stages. Write u, u_1, u_2 for the primitive numerator coordinates, v, v_1 for the corresponding denominator coordinates, and h, h_1 for the two common factors removed in reduction. If the two steps have multipliers a, a_1 , then the exact hypotheses are

$$hu_1 + v = au, \quad hv_1 = av, \quad h_1u_2 + v_1 = a_1u_1.$$

Theorem 3.4 (two-step primitive recurrence). *Under these three equations,*

$$a^2u + hh_1u_2 = h(a + a_1)u_1.$$

Proof. Multiply the first equation by a , replace av using the second equation, and then replace $h_1u_2 + v_1$ using the third. The remaining terms factor as the displayed right-hand side. $\square$

This is the [two-step primitive recurrence](#). The hard point is keeping both changing cancellation factors visible. The result is stronger than a one-step congruence, though it remains non-autonomous and cannot by itself force $a_1 = a^2 - a + 1$.

There is a complementary integer identity when the relevant step is cancellation-free. Put $q = ap - p_1$, and suppose the next primitive coordinate satisfies

$$p_2 + a^2p = (a + a_1)p_1.$$

Then

Theorem 3.5 (cancellation-free curvature identity).

$$q^2 + (pp_2 - p_1^2) = (a_1 - a)pp_1.$$

The square q^2 is the local remainder, while $pp_2 - p_1^2$ is a discrete curvature term. The right side records the change in multiplier, so the identity supplies a diagnostic for an anti-shadowing argument without asserting a sign or a global monotonicity. It is checked as [the cancellation-free curvature square](#); no global anti-shadowing conclusion is claimed.

Finally, the reduced denominator transport has a precise realizability gate. Let e, e_1 be centred errors and let Δ be the Sylvester defect. If

$$hu_1 = u - e, \quad he_1 = a^2e - \Delta(u - e),$$

then the transport equality

$$h((a_1 - 1)u_1 + e_1) = a((a - 1)u + e)$$

has mismatch which factors through $u - e$. Consequently, when $u - e \neq 0$,

$$h((a_1 - 1)u_1 + e_1) = a((a - 1)u + e) \iff a_1 = a^2 - a + 1 + \Delta.$$

This is the exact [feedback transport equivalence](#), obtained from the factored defect at [feedback transport defect](#). The nonzero-row-numerator hypothesis is essential: if $u - e = 0$, the factorised mismatch cannot identify a_1 .

These three identities require orbit-level hypotheses that remain unproved: eventual absence of cancellation, a fixed sign for the curvature, or cofinal passage through the feedback gate. The global mixed-sign and unbounded-negative regimes of Problem #243 remain open.

Theorem 3.6 (sequence form). *Let $a, D, C : \mathbb{N} \rightarrow \mathbb{Z}$ satisfy $D_{n+1} = \text{den}(a_n, D_n)$ and $C_{n+1} = \text{tail}(a_n, D_n, C_n)$. If $E_n = 0$ for all sufficiently large n and $C_{n+1} \neq 0$ for all sufficiently large n , then $a_{n+1} = a_n^2 - a_n + 1$ for all sufficiently large n .*

Formalised as the [eventual Sylvester recurrence](#). The proof is routine: take the larger of the two thresholds and apply Theorem 3.3 at each later index. Every later section is directed at the hypothesis of this theorem, that is, at forcing E to vanish eventually.

The defect identity has a second consequence, which is what makes a single vanishing error worth having.

Theorem 3.7 (zero is absorbing). *Let $a, C, D : \mathbb{N} \rightarrow \mathbb{N}$ be an exact natural orbit, so $C_{n+1} + D_n = a_n C_n$ and $D_{n+1} = a_n D_n$, and let $E_n = \text{ctr}(a_n, D_n, C_n)$. Suppose the centring is strict, $|E_n| < C_n$ for every n . If $E_n = 0$ then $E_{n+1} = 0$.*

Proof. Putting $E_n = 0$ in Theorem 3.1 gives $\Delta_n C_{n+1} = -E_{n+1}$, so C_{n+1} divides E_{n+1} . A multiple of C_{n+1} of absolute value smaller than C_{n+1} is zero, and $|E_{n+1}| < C_{n+1}$ by hypothesis. $\square$

Formalised as the [absorption of a vanishing centred state](#), with the companion [contrapositive along a tail](#): if zero is absorbing beyond some index and E does not vanish eventually, then E is nowhere zero beyond that index.

Absorption changes the shape of the problem. Either E hits zero once and stays there, or it is never zero. Sections 5 and 6 address the special all-negative case only. The mixed-sign analysis in Sections 8 and 9 instead separates cofinally negative errors from an eventually positive tail.

4 Descent when the error is nonnegative

The hypothesis of the theorem below is the update law of Proposition 2.1 read inside $\mathbb{N}$: the error is nonnegative at every index, so the tail state never rises. This is the easy case, and it is settled in three lines.

Theorem 4.1 (descent). *Let $C, E : \mathbb{N} \rightarrow \mathbb{N}$ satisfy $C_{n+1} + E_n = C_n$ for every n . Then $E_n = 0$ for all sufficiently large n .*

Proof. A one-line descent. The relation forces $C_{n+1} \leq C_n$, so C is a nonincreasing sequence of natural numbers and is eventually equal to its minimum; beyond that index $C_{n+1} = C_n$, and the relation gives $E_n = 0$. $\square$

Formalised as the [stabilisation of the nonnegative error](#), on the [eventual constancy of a nonincreasing sequence](#).

The hypothesis is exactly one-sidedness: C and E take values in $\mathbb{N}$. If $E_n < 0$ at infinitely many indices then C rises there, nothing descends, and the argument is unavailable. That is the whole difficulty, and the rest of this note is directed at it.

5 Excluding a constant negative magnitude

Suppose the error is negative with a constant magnitude, $E_n = -m$ for a fixed $m > 0$ and every n . By Proposition 2.1 the state then climbs by exactly m each step, so $C_n = c + nm$ with $c = C_0$, and the definition of the centred state becomes a *shape equation*

$$D_n + m = (a_n - 1)(c + nm). \quad (5.1)$$

Together with $D_{n+1} = a_n D_n$ this is a closed system in (a, D) , and it has no solutions. The following instance shows how the failure happens; the theorem then shows that it cannot be avoided.

Example 5.1 (the shape equation running until it fails). Take $m = c = 1$, so that $C_n = 1 + n$ and the shape equation (5.1) reads $D_n + 1 = (a_n - 1)(1 + n)$, and take $D_0 = 1$. Each step is now determined: at $n = 0$ the equation reads $2 = (a_0 - 1) \cdot 1$, so $a_0 = 3$, and $D_1 = a_0 D_0 = 3$; at $n = 1$ it reads $4 = (a_1 - 1) \cdot 2$, so $a_1 = 3$, and $D_2 = a_1 D_1 = 9$; at $n = 2$ it reads $10 = (a_2 - 1) \cdot 3$, which has no integer solution, and the orbit stops. Longer prefixes occur for other starting values: the finite search recorded in Appendix B found none longer than 17 among the initial states below 5000.

Theorem 5.2 (no constant negative magnitude). *There is no pair of sequences $a, D : \mathbb{N} \rightarrow \mathbb{N}$ with $a_n \geq 2$ for all n satisfying $D_{n+1} = a_n D_n$ and (5.1), for any $m > 0$ and any c . The same holds if the shape only begins at some index.*

Proof when c and m are coprime. Assume first $\gcd(c, m) = 1$. Two facts about the multipliers do all the work, and the difficulty lies in the fact that they concern the same finite set of primes from opposite directions: the first makes every multiplier meet that set, the second lets each prime of the set meet at most one multiplier.

Every a_j shares a prime with m . Suppose $\gcd(a_j, m) = 1$. Then m is invertible modulo a_j , so some n has $c + nm \equiv 0$, that is $a_j \mid C_n$; and $a_j \mid D_n$ for every $n > j$, since D is multiplicative with a_j among its factors. Choosing such an n beyond j and reading (5.1) modulo a_j gives $a_j \mid m$, contradicting $\gcd(a_j, m) = 1$ and $a_j \geq 2$.

A prime divisor of m occurs in at most one a_j . Suppose $p \mid m$ and $p \mid a_j$. For $n > j$ we have $p \mid D_n$, so (5.1) gives $(a_n - 1)C_n \equiv 0 \pmod{p}$. Now $C_n = c + nm \equiv c$, and $p \nmid c$ because $p \mid m$ and $\gcd(c, m) = 1$; hence $p \mid a_n - 1$, so $p \nmid a_n$. Thus p cannot divide any later multiplier.

The two facts are incompatible. Each of the infinitely many multipliers needs a prime divisor of m , and each prime divisor of m serves at most one multiplier, while m has only finitely many prime divisors. $\square$

The case $\gcd(c, m) = 1$ is the [scale-primitive exclusion](#). The special case $m = c = 1$, worked through in Example 5.1, where m has no prime divisors at all and the first fact is immediately contradictory, is recorded separately as the [normalised exclusion](#).

Removing the scale. For general c , put $g = \gcd(c, m)$. Equation (5.1) shows $g \mid D_n$ for every n , so dividing D , c and m by g leaves a system of the same shape with coprime data, which the previous case excludes. $\square$

Formalised as the [exclusion at every scale](#), and the eventual form, obtained by shifting the orbit to the first constant index, as the [eventual exclusion](#).

Theorem 5.2 excludes a centred state that is eventually constant and negative, at every magnitude and every scale. It says nothing about a negative state whose magnitude changes, and the argument genuinely uses constancy: the finiteness of the set of prime divisors of m is what the pigeonhole uses, and a varying magnitude presents a fresh set at each index.

6 Excluding a periodic negative magnitude

The next case allows the magnitude to vary, provided it repeats. Write $e_n = -E_n > 0$ for the magnitude, as in Section 2, so that $C_{n+1} = C_n + e_n$; suppose e has period h , meaning $e_{n+h} = e_n$ for every n , and that the state gains a fixed *drift* $M > 0$ over one period, meaning $C_{n+h} = C_n + M$ for every n .

At $h = 1$ this says that the magnitude is constant and that M is its value, which is the situation of Section 5; Theorem 5.2 already excludes it, and does so without the extra hypothesis $e_n < a_n$ imposed below. The new content is $h \geq 2$.

Section 5 could run its pigeonhole on the prime divisors of the single number m . Here the magnitude changes inside a period, and the number available to run it on is the drift M . Three lemmas replace the two facts of Section 5. The first is that multipliers persist: every a_j divides every strictly later denominator state, the [persistence of a multiplier](#). The second is a *lock*: a prime already dividing D and also dividing the current multiplier is forced into the next tail state, the [one-step lock](#), and once present in both D and C it stays in every later D , C and magnitude, the [persistence of a lock](#). The third transports a lock backwards through the period: if a divisor of the drift M occurs in two multipliers, it is a common divisor of C_0 and of every magnitude, the [repeated-divisor transport](#), using that both the period relation and the drift iterate over any number of periods, the [iterated period](#) and the [iterated drift](#).

Theorem 6.1 (no periodic negative magnitude). *Let $a, D, C, e : \mathbb{N} \rightarrow \mathbb{N}$ with $a_n \geq 2$, $e_n > 0$ and $e_n < a_n$ for every n , satisfying*

$$D_{n+1} = a_n D_n, \quad C_{n+1} = C_n + e_n, \quad D_n + e_n = (a_n - 1)C_n,$$

and suppose $e_{n+h} = e_n$ and $C_{n+h} = C_n + M$ for some $h > 0$ and $M > 0$. This is impossible.

Proof. Strong induction on the drift M . The difficulty lies in the fact that a repeated prime divisor of M need not contradict anything directly. It instead forces a common scale on the whole orbit, which can be divided out, and the induction runs on the drift that the division reduces.

Suppose first that no prime divisor of M is a common divisor of C_0 and of every magnitude. Repeated-divisor transport then applies in the contrapositive: a prime divisor of M occurring in two of the multipliers would be exactly such a common divisor, so each prime divisor of M occurs in at most one multiplier. Against this, the argument of Section 5 run phase by phase (along a fixed residue class modulo h the magnitude is constant and C advances by M at each period, so the multipliers of that phase meet an arithmetic progression exactly as in Section 5) makes every multiplier share a prime with M . Since M has only finitely many prime divisors and there are infinitely many multipliers, pigeonhole gives a contradiction.

Otherwise some prime p divides M , divides C_0 , and divides every magnitude. Then $p \mid C_n$ for every n , since $C_{n+1} = C_n + e_n$ and both summands on the right are divisible by p , and the shape equation $D_n + e_n = (a_n - 1)C_n$ then gives $p \mid D_n$ for every n . Divide D , C and e by p . The multipliers are untouched, so $a_n \geq 2$ and $e_n < a_n$ persist and the magnitudes stay positive; the three recurrences are homogeneous in (D, C, e) and so

survive; the period is still h ; and the drift becomes $M/p < M$. The inductive hypothesis applies. $\square$

The first of the two cases above is the [phase-primitive exclusion](#), the induction is the [exclusion at every scale](#), and the eventual form is the [eventual exclusion](#).

The bound $e_n < a_n$ is a genuine restriction and not a normalisation. It is the range in which a multiplier cannot divide its own phase magnitude, which is what the lock argument needs. It holds in the intended reading, where e_n is a centred representative and a_n grows quadratically, but it is assumed here and not derived. The drift hypothesis $M > 0$ is also used: a periodic magnitude with zero drift would be identically zero, which is the case already settled by Section 4.

Remark. On the canonical orbit of [7], read through the dictionary of Section 1, the bound $e_n < a_n$ holds eventually. Lemma 15(2) there gives the centring range $-C_n/2 \leq E_n < C_n/2$, so $e_n = |E_n| \leq C_n/2$ at a negative index, and the proof of Lemma 15(3) gives $C_{n+1} \leq \frac{3}{2}C_n$ [7, p. 9], hence the uniform bound $C_n \leq C_0(3/2)^n$. Against this, summability of $\sum 1/a_n$ forces $a_n \rightarrow \infty$, and the growth hypothesis $a_n^2/a_{n+1} \rightarrow 1$ gives $a_{n+1} \geq a_n^2/2$ for all large n , hence $a_{n+1} \geq 2a_n$ once $a_n \geq 4$; so the multipliers grow at least geometrically with ratio 2 and outgrow the bound on C_n , making e_n/a_n eventually small. Two caveats. Summability is used for this step, so the estimate is not available from the state recurrence alone; and the conclusion is eventual and not universal, whereas Theorem 6.1 imposes $e_n < a_n$ at every index, so it is the eventual form of the exclusion cited above that applies. Under the periodicity hypothesis of Theorem 6.1 the estimate is in any case unnecessary, since a periodic magnitude is bounded while the multipliers tend to infinity.

7 Bounded rise cannot remain coprime to fresh moduli

Periodicity is still a strong assumption. Removing it needs a different kind of obstruction, and the one used here is a counting argument about where an integer sequence with bounded upward steps must land.

Lemma 7.1 (shifted blocks of consecutive multiples). *Let $m_0, \dots, m_{B-1}$ be pairwise coprime and at least 2. For every bound there is a t beyond it with $m_i \mid t + i$ for each $i < B$.*

Proof. Standard Chinese remaindering: solve $t \equiv -i \pmod{m_i}$ simultaneously, then add multiples of $\prod_i m_i$ to pass the bound. $\square$

Formalised as the [shifted consecutive multiples](#).

Example 7.2 (a block of three consecutive multiples). Take $B = 3$ and $(m_0, m_1, m_2) = (3, 4, 5)$. The congruences $t \equiv 0 \pmod{3}$, $t \equiv 3 \pmod{4}$ and $t \equiv 3 \pmod{5}$ hold exactly when $t \equiv 3 \pmod{60}$. At $t = 3$ the block is 3, 4, 5 with $3 \mid 3$, $4 \mid 4$ and $5 \mid 5$; at $t = 63$ it is 63, 64, 65 with $3 \mid 63$, $4 \mid 64$ and $5 \mid 65$. A sequence of natural numbers that starts at 0, tends to infinity and rises by at most 3 at each step cannot step over the block $\{3, 4, 5\}$: it has a first value at least 3, that value is at most 5, and every member of $\{3, 4, 5\}$ is divisible by one of the three moduli.

Theorem 7.3 (bounded rise cannot remain coprime to fresh moduli). *Let $u : \mathbb{N} \rightarrow \mathbb{N}$ tend to infinity with $u_{n+1} \leq u_n + B$ for a fixed $B > 0$. Then u cannot remain coprime to infinitely many fresh pairwise coprime moduli: there is no family of pairwise coprime $m_i \geq 2$, one for each index, such that $\gcd(m_i, u_i) = 1$ whenever $i < t$.*

Proof. Take B of the moduli, from indices beyond any chosen point, and use Lemma 7.1 to find t far out with the i th of them dividing $t + i$. The interval $[t, t + B)$ has length B , and u climbs by at most B per step while tending to infinity, so some u_s lands in it. Then $u_s = t + i$ for some $i < B$, and the i th modulus divides u_s . Since that modulus is at least 2, this contradicts $\gcd(m_i, u_i) = 1$. $\square$

Formalised as the **bounded-rise barrier**. The key point is that the two hypotheses pull against each other exactly: divergence forces u to travel arbitrarily far, and bounded rise forbids it from stepping over a window of width B . Uniformity of the bound is what the proof uses, and it is the hypothesis that cannot be relaxed by this argument: a rise bounded only along a subsequence leaves the intervening steps free to jump the window. We have not looked for a sequence of unbounded rise remaining coprime to every fresh modulus. The statement is elementary and we are not aware of a prior published form of it, but we would not be surprised if one exists.

The moduli are indexed by the same set as the sequence, and the i th of them is asked to divide no value u_t with $t > i$; nothing at all is asked of it at or before its own index. That asymmetry is what makes the hypothesis available in the application below, where m_i is the i th multiplier and only the later numerators are known to be coprime to it.

The barrier applies to the problem because a *reduced* exact tail already carries pairwise coprime moduli. Call (u, v) a reduced exact tail with multipliers a when $\gcd(u_n, v_n) = 1$,

$$u_{n+1} + v_n = a_n u_n, \quad v_{n+1} = a_n v_n.$$

Here u is the numerator and v the denominator: the two recurrences are those of Section 2, with u in the role of the tail state C and v in that of the denominator state D , and with coprimality imposed at every index.

Proposition 7.4 (reduced tails are pairwise coprime). *In a reduced exact tail, a_n is coprime to v_n ; the multipliers at distinct indices are pairwise coprime; and every earlier multiplier is coprime to every later numerator.*

These are the **step coprimality**, the **pairwise coprimality**, and the **whole-modulus avoidance**. Feeding them to Theorem 7.3 gives the form used later: there is no reduced exact tail whose numerator tends to infinity with a uniformly bounded upward increment, the **reduced bounded-rise exclusion**, together with its eventual form, the **eventual reduced exclusion**.

Example 7.5 (Sylvester's sequence as a reduced exact tail). Put $u_n = 1$ and $v_n = D_n = 1, 2, 6, 42, 1806, \dots$ as in Example 2.2, with multipliers $a_n = v_n + 1 = 2, 3, 7, 43, 1807, \dots$. Then $\gcd(u_n, v_n) = 1$, $u_{n+1} + v_n = 1 + v_n = a_n u_n$ and $v_{n+1} = a_n v_n$, so this is a reduced exact tail, and Proposition 7.4 returns the classical fact that Sylvester's numbers are pairwise coprime. Its numerator is constant, so it satisfies every hypothesis of the exclusion just stated except divergence; since the tail exists, that hypothesis cannot be dropped.

An arbitrary orbit of Section 2 need not be reduced, so one more step is needed before the barrier can be applied to it: the common factor must stop changing. We call a set of indices *cofinal* when it is infinite, and say that a property holds *cofinally* when the set of indices at which it holds is cofinal. The step below is the only one connecting the orbit of Section 2 to the reduced tails of Proposition 7.4, and it is where the cofinal boundedness in the proof of Theorem 8.1 is used.

Proposition 7.6 (the tail gcd stabilises). *Let (a, D, C) be an exact natural orbit, that is, a triple of $\mathbb{N}$ -valued sequences with $C_{n+1} + D_n = a_n C_n$ and $D_{n+1} = a_n D_n$, whose centred state $E_n = \text{ctr}(a_n, D_n, C_n)$ is negative cofinally, with magnitudes bounded along that cofinal set. Then $\gcd(C_n, D_n)$ is eventually constant, and beyond that index the orbit divided by the stable gcd is a reduced exact tail.*

Proof. The tail gcd $\gcd(C_n, D_n)$ divides its successor, the [gcd monotonicity](#), so the tail gcds form a positive divisibility chain. At an index where the centred state is negative the tail gcd is exactly $\gcd(C_n, e_n)$, the [gcd at a negative index](#), and is therefore at most the magnitude there; so the chain is bounded along the cofinal set of negative indices. A positive divisibility chain whose values are bounded along a cofinal set is eventually constant, the [chain stabilisation](#). Hence cofinally bounded negative magnitudes make the tail gcd stabilise, the [gcd stabilisation](#), and beyond that point dividing by the stable gcd leaves the numerator and denominator coprime, that is, a reduced exact tail. $\square$

The stabilisation of the tail gcd is the checked statement cited at the end of the proof; the last clause, that the divided orbit is a reduced exact tail, is exposition and carries no separate label. Boundedness is required only *along a cofinal set*. That is what the situation gives: the tail gcd is controlled only at the negative indices, and cofinally many of them is all the divisibility chain needs.

Normalised vanishing by itself gives a different, strictly weaker conclusion. For an exact natural orbit with $C_n > 0$, put

$$G_n = \gcd(C_n, D_n), \quad \Gamma(N) = \#\{0 \leq j < N : G_j < G_{j+1}\}.$$

Proposition 7.7 (normalised vanishing makes strict gcd changes sparse). *Assume*

$$(\forall K \geq 1)(\exists N)(\forall n \geq N) \quad K|E_n| < C_n.$$

Then, for every $K \geq 1$, eventually $K\Gamma(N) < N$. Moreover, for every starting bound B and block length L , some $n \geq B$ satisfies

$$G_n = G_{n+1} = \dots = G_{n+L}.$$

The first assertion is the [sublinear strict-growth theorem](#); the second is the [arbitrarily late constant-block theorem](#). The proof first converts normalised vanishing into subexponential growth of C_n , then charges every strict divisibility increase of G_n against a finite power-of-two budget. Sparse strict changes force long finite gaps between them.

The quantifiers matter. Proposition 7.6 uses cofinally bounded negative magnitudes and yields eventual constancy. Proposition 7.7 uses only normalised vanishing and yields arbitrarily late constant blocks of each prescribed finite length. It does not yield one infinite constant tail, and it does not exclude cofinally unbounded negative excursions.

8 Excluding a bounded negative part

Assembling the previous section gives Theorem 8.1, the endpoint of the barrier argument. Theorem 9.1 in the next section uses a different finiteness hypothesis. The two results are incomparable: Theorem 9.1 assumes only the state recurrence with $C_n > 0$, while hypothesis (5) below does not imply finite *normalised negative mass*, by which we mean convergence of $\sum_n (-E_n)_+ / C_n$, where $x_+ = \max(x, 0)$, since a constant negative part $-b$ gives $C_n \sim bn$ and hence a divergent sum. The hypotheses of Theorem 8.1 are listed in full, because two of them are analytic inputs that are assumed here rather than derived from the growth condition of Problem 1.1.

Informally, hypotheses (5) and (6) below pull against each other. By Proposition 2.1, hypothesis (5) caps the rise of the tail state at B per step, while hypothesis (6), once the error is nowhere zero, forces the tail state to diverge. Proposition 7.6 makes the orbit reduced, and Theorem 7.3 then converts the tension into a contradiction: an integer sequence tending to infinity with upward steps of size at most B cannot remain coprime to the pairwise coprime moduli a reduced tail carries. The complementary case, in which the error is eventually positive, is the descent of Theorem 4.1.

Theorem 8.1 (bounded negative part). *Let $a, C, D : \mathbb{N} \rightarrow \mathbb{N}$ and $E : \mathbb{N} \rightarrow \mathbb{Z}$ satisfy*

1. $a_n > 1$ and $C_n > 0$ for every n ;
2. the exact dynamics $C_{n+1} + D_n = a_n C_n$ and $D_{n+1} = a_n D_n$;
3. $E_n = \text{ctr}(a_n, D_n, C_n)$ for every n ;
4. eventual strict centring: $|E_n| < C_n$ for all large n ;
5. eventually bounded negative part: $-B \leq E_n$ for all large n , for some B ;
6. normalised vanishing: for every K there is an N with $K|E_n| < C_n$ for all $n \geq N$.

Then $E_n = 0$ for all sufficiently large n .

Proof. Suppose not. We first remove every late zero: by Theorem 3.7 and its contrapositive form, E is nowhere zero beyond the centring threshold. Two cases remain. If E is negative cofinally, then (5) bounds those magnitudes, so Proposition 7.6 makes the tail gcd stabilise and the orbit may be taken reduced past that point. Hypothesis (5) with Proposition 2.1 gives $C_{n+1} \leq C_n + B$, a bounded rise; hypothesis (6) with E nowhere zero gives $C_n \rightarrow \infty$. A reduced exact tail with a divergent numerator and bounded rise is excluded by Theorem 7.3. It remains to treat the case that E is eventually positive, and there the descent of Theorem 4.1 applies and forces E to vanish, contradicting nowhere-vanishing. $\square$

Formalised as the [bounded-negative-part rigidity](#), with the eventual form, in which the centring and the bound need only hold from some index, as the [eventual bounded-negative-part rigidity](#). The two auxiliary results it uses are the [divergence from normalised vanishing](#) and the [exclusion of cofinally bounded negative magnitudes](#), the latter over the [tail-divergence form](#).

Corollary 8.2. *Under the hypotheses of Theorem 8.1, together with $C_{n+1} \neq 0$ for all large n , the multipliers satisfy $a_{n+1} = a_n^2 - a_n + 1$ for all sufficiently large n .*

Proof. Theorem 8.1 then Theorem 3.6. $\square$

Hypotheses (1)–(3) are the state system, and (6) is the division-free form of $|E_n|/C_n \rightarrow 0$. Hypothesis (4) is logically redundant: hypothesis (6) with $K = 1$ gives $|E_n| < C_n$ eventually. It is retained because the displayed statement tracks the linked Lean declaration; shifting to the resulting threshold gives the same conclusion without an independent centring input. The formal source does not derive normalised vanishing from $a_{n+1}/a_n^2 \rightarrow 1$, so Theorem 8.1 is, by itself, a conditional theorem about the state system. Combined with Koizumi’s bridge below, however, Corollary 8.2 settles the bounded-negative canonical case of Problem 1.1. What is unconditional in this note is the all-negative constant and periodic exclusions, Theorems 5.2 and 6.1, and the barrier itself, Theorem 7.3.

The conditionality can nevertheless be located precisely, because the missing inputs are supplied elsewhere. With the results of [7], Theorem 8.1 becomes a conditional theorem about Problem 1.1 itself rather than about the state system alone. Let (a_n) satisfy the hypotheses of Problem 1.1. After deleting a finite prefix, Corollary 10 of [7, p. 8] makes the sequence the pseudo-greedy expansion of its own reciprocal sum, and Lemma 15 there supplies the integers c_n, d_n, e_n of Section 1. Hypothesis (1) then holds: c_n is a positive integer by Lemma 15(1), and $a_n > 1$ after a further finite shift, because summability of $\sum 1/a_n$ forces $a_n \rightarrow \infty$. Hypothesis (2) is the pair of recurrences $c_{n+1} = a_n c_n - d_n$ and $d_{n+1} = a_n d_n$ of Lemma 15(2), and hypothesis (3) is that lemma’s $a_n = (d_n - e_n)/c_n + 1$ read as $e_n = d_n - (a_n - 1)c_n$, which is the map ctr. The centring range $-c_n/2 \leq e_n < c_n/2$ in the same lemma gives hypothesis (4) at every index, which is stronger than the eventual form used here. Hypothesis (6) is the vanishing of the gap sequence in Corollary 10, since $\varepsilon_n = e_n/c_n$. The sole hypothesis not supplied is (5), the eventual bound on the negative part; nothing in [7] yields it and nothing here does either, so Problem #243 stays open.

9 Excluding finite normalised negative mass

Theorem 8.1 uses a uniform bound on the negative part. A different finiteness hypothesis, on the total normalised negative mass, removes a further family of orbits, and the argument is multiplicative rather than arithmetic: it compares C with a convergent product instead of with a system of congruences.

Theorem 9.1 (finite normalised negative mass). *Let $C_n \in \mathbb{N}_{>0}$ and $E_n \in \mathbb{Z}$ satisfy*

$$C_{n+1} = C_n - E_n, \quad \forall K \exists N \forall n \geq N, \quad K|E_n| < C_n.$$

If

$$\sum_{n=0}^{\infty} \frac{(-E_n)_+}{C_n} < \infty,$$

then $E_n = 0$ eventually. For an exact reciprocal-tail orbit this implies $a_{n+1} = a_n^2 - a_n + 1$ eventually.

Proof. Put $\delta_n = (-E_n)_+/C_n$. The update gives $C_{n+1} \leq C_n(1 + \delta_n)$, so

$$C_N \leq C_0 \prod_{n < N} (1 + \delta_n).$$

The product is bounded because $\sum \delta_n$ converges. Choose an integer M above that bound and apply normalised vanishing with $K = M$. Then $M|E_n| < C_n < M$ for all sufficiently large n , forcing the integer E_n to be zero. The final recurrence is Theorem 3.6, whose hypothesis $C_{n+1} \neq 0$ holds because $C_n > 0$ throughout. $\square$

The product argument is the [summable relative growth](#); its specialisation to negative mass is the [vanishing from finite negative mass](#), and the statement giving the recurrence directly is the [Sylvester recurrence from summable negative mass](#). This is a checked implication about the state system, not a derivation of normalised vanishing from Problem #243. Normalised vanishing is still an input: it is what converts the bound on C into a bound on $|E_n|$, and nothing here derives it from the growth condition of Problem 1.1.

The transfer of the previous section applies here as well. On the canonical orbit of Corollary 10 of [7, p. 8] the recurrence $C_{n+1} = C_n - E_n$ holds with $C_n > 0$ by Lemma 15 [7, p. 9], and normalised vanishing is the vanishing of the gap sequence, so Theorem 9.1 is a conditional theorem about Problem 1.1 whose sole remaining hypothesis is summability of $\sum_n (-E_n)_+/C_n$. That hypothesis is not supplied there, and is the content of Problem 11.5 below.

Example 9.2 (the product bound at a geometric rate). Suppose $C_0 = 100$ and $(-E_n)_+/C_n \leq 2^{-n-1}$ for every n , so that the normalised negative mass is at most 1; the constraint at $n = 0$ still permits E_0 as negative as -50 . Since $1 + x \leq e^x$,

$$C_N \leq 100 \prod_{n < N} (1 + 2^{-n-1}) \leq 100e < 272$$

for every N . Normalised vanishing at $K = 272$ then gives $272|E_n| < C_n < 272$ for all large n , so $|E_n| < 1$ and $E_n = 0$ there. The constant 272 comes from the total mass and not from any single magnitude, which is why the hypothesis of Theorem 9.1 is summability of $(-E_n)_+/C_n$ and not a bound on it.

10 Weighted first crossings of LCM records

The summability criterion admits a smaller series: only steps setting a new LCM numerator record need contribute, a fixed LCM-scale baseline may be subtracted, and the remaining mass may be logarithmically discounted. The result in this section is an ordinary mathematical proof. The [finite arithmetic and charging source](#) is available separately; the global analytic argument is not formalised.

Set

$$L_n = \text{lcm}(q, a_0, \dots, a_{n-1}), \quad M_n = D_n/L_n, \quad U_n = C_n/M_n, \quad V_n = E_n/M_n.$$

The rational tail has denominator dividing L_n , so U_n and V_n are integers. With $\rho_n = \text{gcd}(L_n, a_n)$, the exact updates are

$$M_{n+1} = M_n \rho_n, \quad \rho_n U_{n+1} = U_n - V_n, \quad V_n = L_n - (a_n - 1)U_n.$$

Write $R_n = \max_{j \leq n} U_j$ and $\mathcal{R} = \{n : U_{n+1} > R_n\}$. Centring implies that every sufficiently late strict rise has $\rho_n = 1$: otherwise $U_{n+1} \leq 3U_n/4$. Its actual jump is therefore $d_n = U_{n+1} - U_n = -V_n$.

Theorem 10.1 (weighted record excess). *Assume the growth and rationality hypotheses of Problem 1.1. Let $f : [1, \infty) \rightarrow [0, \infty)$ be finite and nonincreasing, with $\int_1^\infty f(t) dt = \infty$. Then the sequence is eventually Sylvester if and only if, for some integer $B \geq 0$,*

$$\sum_{n \in \mathcal{R}} (-V_n - B)_+ f(U_n) < \infty.$$

Proof. Fix $B \geq 1$. There are eventually B pairwise coprime old digits $m_0, \dots, m_{B-1} > B$ dividing L_T . Indeed, $C_{n+1}/C_n = 1 - E_n/C_n \rightarrow 1$ gives $\log C_n = o(n)$; since $M_n \leq C_n$, the update for M_n permits only $o(n)$ indices with $\rho_n > 1$. The other digits are mutually coprime and tend to infinity.

Put $P = \prod_i m_i$ and choose x by the Chinese remainder theorem with $m_i \mid x + i$. Consider the heights $\tau = x + B + kP$. A first crossing $U_n \leq R_n < \tau \leq U_n + d_n$ is a record step. If $d_n \leq B$, then $U_n \in [\tau - B, \tau)$, so some m_i divides U_n . It also divides L_n , hence divides $d_n = (a_n - 1)U_n - L_n$, contradicting $0 < d_n \leq B < m_i$.

If the step first crosses $h \geq 1$ such heights, their spacing gives $(h - 1)P < d_n$. With $r = d_n - B \geq 1$ and $P \geq B + 1$, we have $d_n = B + r \leq Pr$, whence $h \leq r$. Monotonicity of f now gives

$$\sum_{\substack{\tau \text{ first crossed} \\ \text{at step } n}} f(\tau) \leq (d_n - B)f(U_n).$$

Each height above R_T has one first crossing. Summing, and comparing each interval of length P with its left endpoint, yields the finite bound

$$\sum_{\substack{T \leq n < N \\ n \in \mathcal{R}}} (-V_n - B)_+ f(U_n) \geq \frac{1}{P} \int_{R_T+P}^{R_N} f(t) dt. \quad (1)$$

If the endpoint fails, centred zero is never reached on a sufficiently late tail. Integrality gives $1/U_n \leq |V_n|/U_n = |E_n|/C_n \rightarrow 0$, hence $R_n \rightarrow \infty$. The right side diverges, for every $B \geq 1$; $B = 0$ follows by domination. Conversely a Sylvester tail telescopes to $x_n = 1/(a_n - 1)$, so $V_n = 0$ eventually. $\square$

For example, $f(t) = 1/[t \log(et)]$ gives the sufficient condition

$$\sum_{n \in \mathcal{R}} \frac{(-V_n - B)_+}{U_n \log(eU_n)} < \infty.$$

Its finite lower bound in (1) is $P^{-1} \log(\log(eR_N)/\log(e(R_T + P)))$. Further fixed iterated logarithmic factors are allowed whenever the integral still diverges. These are specialisations of one crossing theorem.

The criterion also has an exact expression in the original growth defect. Put $\gamma_n = a_n^2/a_{n+1} - 1$ and $\theta_n = E_n/C_n$. The defect identity gives

$$\gamma_n + \theta_n = \frac{(1 - \theta_n)(a_n - 1 + \theta_{n+1})}{a_{n+1}}, \quad 0 < \gamma_n + \theta_n < 3/a_n$$

eventually. Thus the two nonnegative summands $U_n f(U_n)(\gamma_n - B/U_n)_+$ and $(-V_n - B)_+ f(U_n)$ differ by at most $3U_n f(U_n)/a_n$. This is summable because $U_n \leq C_n = \exp(o(n))$, $f(U_n) \leq f(1)$, and $a_n \geq \exp(c2^n)$ eventually for some $c > 0$. Consequently Theorem 10.1 is equivalent to finiteness of

$$\sum_{n \in \mathcal{R}} U_n f(U_n) \left(\frac{a_n^2}{a_{n+1}} - 1 - \frac{B}{U_n} \right)_+ \quad (2)$$

for some B . The original hypotheses do not currently supply this finiteness. In particular, termwise convergence to zero is insufficient. Also, d_n is the actual jump, including any recovery from a drawdown; it must not be replaced by $R_{n+1} - R_n$ in the crossing proof.

11 Complements and further questions

Problem #243 is open. The negative case has narrowed, and it is worth being precise about what is left.

Each exclusion holds only in its stated regime. Theorems 5.2 and 6.1 exclude the constant and the stated periodic magnitudes on an all-negative tail, and do not exclude those patterns merely along the negative indices of a mixed-sign tail; Theorems 8.1 and 9.1 exclude a bounded negative part and finite normalised negative mass under normalised vanishing. Together with absorption and descent they fix the profile of any counterexample.

Proposition 11.1 (frontier profile for a counterexample). *For the canonical integer state attached to any counterexample to Problem 1.1,*

$$E_n \neq 0 \text{ eventually, } \frac{|E_n|}{C_n} \rightarrow 0,$$

and

$$\limsup_{\substack{n \rightarrow \infty \\ E_n < 0}} (-E_n) = \infty, \quad \sum_{n=0}^{\infty} \frac{(-E_n)_+}{C_n} = \infty.$$

In particular, negative indices occur infinitely often. Thus the remaining regime consists of globally coherent, unbounded negative excursions with divergent normalised negative mass.

Proof. Koizumi's canonical-tail transfer gives strict centring and normalised vanishing after a finite shift; the strict-centring hypothesis of Theorem 8.1 also follows from normalised vanishing with $K = 1$. Absorption then makes $E_n \neq 0$ eventually, since eventual zero would give the Sylvester recurrence. Descent excludes an eventually nonnegative state. Theorem 8.1 excludes a bounded negative part, and Theorem 9.1 excludes finite normalised negative mass. These statements are unchanged by deleting a finite prefix. $\square$

Put

$$\delta_n := \frac{(-E_n)_+}{C_n}.$$

Every argument in Sections 5–8 uses a finiteness hypothesis: a fixed set of prime divisors, a fixed period, or a fixed bound on the negative part. Theorem 9.1 uses $\sum_n \delta_n < \infty$.

Proposition 11.1 records that a counterexample has none of them; it is a frontier statement, not an additional problem equivalent to the original one. Stated at the level of state orbits, the surviving obstruction is the following.

Problem 11.2 (unbounded divergent-mass negative excursions). Exclude, or construct, an exact natural orbit (a, D, C) with $a_n > 1$ and $C_n > 0$, whose multipliers satisfy $\lim_n a_{n+1}/a_n^2 = 1$ and whose centred state satisfies normalised vanishing, and which is negative infinitely often with magnitudes unbounded along that cofinal set and

$$\sum_n \frac{(-E_n)_+}{C_n} = \infty.$$

An excursion of this kind has none of the finiteness hypotheses just listed. By Proposition 11.1 the canonical state of a counterexample is such an orbit after deletion of a finite prefix, so an exclusion would settle Problem #243. A construction would not settle it in the other direction: an exact state orbit need not arise from a sequence satisfying Problem 1.1.

Remark. The hypotheses in Problem 11.2 beyond the state recurrence are not decorative: the recurrence alone admits unbounded divergent-mass excursions. Put

$$C_n = 2^n, \quad b_0 = 2, \quad b_{n+1} = \frac{1}{2}b_n(b_n + 2), \quad a_n = b_n + 2, \quad D_n = b_n C_n,$$

so that $(b_n) = 2, 4, 12, 84, 3612, \dots$ and $(a_n) = 4, 6, 14, 86, 3614, \dots$. Every b_n is a positive even integer, since $b_n = 2k$ gives $b_{n+1} = 2k(k + 1)$, so all four sequences are $\mathbb{N}$ -valued. The orbit is exact: $C_{n+1} + D_n = 2^n(2 + b_n) = a_n C_n$ and $D_{n+1} = b_{n+1} 2^{n+1} = b_n(b_n + 2)2^n = a_n D_n$. Its centred state is

$$E_n = D_n - (a_n - 1)C_n = b_n 2^n - (b_n + 1)2^n = -C_n,$$

so the negative magnitudes 2^n are unbounded and $\sum_n (-E_n)_+/C_n$ diverges. What fails is the centring: $|E_n| = C_n$ at every index, so strict centring fails at the boundary and normalised vanishing fails outright. The multipliers satisfy $a_{n+1} = \frac{1}{2}(a_n^2 - 2a_n + 4)$, so $a_{n+1}/a_n^2 \rightarrow \frac{1}{2}$ and the growth hypothesis fails as well. The example therefore says nothing about Problem 1.1, and it does not isolate the centring hypotheses: $E_n = -C_n$ at every index forces that multiplier recurrence, so growth fails with them. It is recorded only to show that the state recurrence alone is not enough.

11.1 Global height and old-factor overlap

The main arithmetic gap is global. Define the cumulative digit LCM and its overlap quotient by

$$L_0 = D_0, \quad L_{n+1} = \text{lcm}(L_n, a_n), \quad M_n = \frac{D_n}{L_n}.$$

Since $D_n = D_0 \prod_{j < n} a_j$, the quotient is integral and $M_n L_n = D_n$. It records the multiplicity lost when the full product scale is compressed to an LCM.

11.2 Repair entropy across a recovery

The cumulative quotient does not by itself distinguish one large deletion from many small ones. The reduced recurrence does. Let $u, h, c : \mathbb{N} \rightarrow \mathbb{N}$ and $e : \mathbb{N} \rightarrow \mathbb{Z}$ satisfy

$$h_n u_{n+1} = u_n - e_n, \quad u_n > 0, \quad h_n > 0.$$

Fix $K, r, L \in \mathbb{N}$ with $K, L > 0$, assume

$$K|e_{r+i}| < u_{r+i} \quad (0 \leq i < L), \quad u_r \leq u_{r+L},$$

and let R be a finite family of moduli. Suppose every $m_q, q \in R$, divides the deletion product $\prod_{r \leq n < r+L} c_n$, and that $c_n^2 \mid h_n$ throughout the interval. Then

$$K^L \text{lcm}(m_q : q \in R)^2 < (K+1)^L. \quad (3)$$

This is the [repair-entropy conservation law](#).

The square in (3) is the key point. Exact denominator reduction charges c_n^2 to the scale growth, so a repaired family whose LCM is at least $2^{|R|}$ satisfies

$$K^L 4^{|R|} < (K+1)^L.$$

The formal consequence is [the independent-repair cardinality bound](#). It rules out positive-density deletion of independent old moduli inside recoveries with a fixed relative-error budget.

There is also an endpoint that does not mention a chosen modulus family. If $K|e_n| < u_n$ eventually for every K , then for each fixed $L > 0$ there is an N such that every recovery $u_r \leq u_{r+L}$ with $r \geq N$ satisfies

$$\prod_{0 \leq i < L} h_{r+i} = 1.$$

Thus a surviving nontrivial reset must have recovery length tending to infinity. Lean checks this as [eventual disappearance of fixed-length reset payments](#). The theorem does not bound a recovery length that varies with r , nor does it supply a lower bound for the LCM of the moduli repaired in such an interval. Those are the missing inputs needed to turn (3) into a global contradiction.

Problem 11.3 (global overlap-height growth). For every nonterminal canonical orbit satisfying Problem 1.1, must

$$\limsup_{n \rightarrow \infty} \frac{\log M_n}{n} > 0?$$

Equivalently, must there be a $K \geq 1$ for which

$$2^n \leq M_n^K$$

at infinitely many indices?

The two displayed formulations are equivalent up to changing the positive constant; the second is not a weaker target. On the canonical orbit, Section 10 supplies $M_n \mid C_n$ and $M_n \leq C_n$. Therefore a positive answer would contradict the subexponential tail-height budget and prove Problem #243. The missing input is the proposed lower bound on overlap growth.

There is also a more local-looking question whose content is nevertheless the entire prefix. Put $A_n = \prod_{j < n} a_j$, so $D_n = D_0 A_n$. From

$$D_0 A_n = (a_n - 1)C_n + E_n$$

one immediately obtains

$$\gcd(A_n, a_n - 1) \mid E_n.$$

The checked [tail-height estimate](#) and normalised vanishing make $|E_n|$ subexponential in n .

Problem 11.4 (old-factor overlap). In every nonterminal canonical orbit satisfying Problem 1.1, is

$$\limsup_{n \rightarrow \infty} \frac{\log \gcd(A_n, a_n - 1)}{n} > 0?$$

Equivalently, do there exist $\eta > 0$ and infinitely many n such that $\gcd(A_n, a_n - 1) \geq e^{\eta n}$?

A positive answer contradicts the displayed divisibility and the subexponential error budget. Arbitrarily long locally admissible blocks do not answer this question: the gcd uses the complete prefix.

11.3 The direct analytic question

Problem 11.5 (summability from rationality). Let (a_n) satisfy Problem 1.1, and let (D_n, C_n, E_n) be its canonical integer state. Must

$$\sum_{n=0}^{\infty} \delta_n = \sum_{n=0}^{\infty} \frac{(-E_n)_+}{C_n} < \infty?$$

An affirmative answer closes Problem #243 by Theorem 9.1. Since $\delta_n \rightarrow 0$, convergence is equivalently expressible as boundedness of the partial products $\prod_{n < N} (1 + \delta_n)$; this is the same criterion in multiplicative form. A negative answer requires a sequence satisfying the full growth and rationality hypotheses. A locally admissible state orbit is insufficient. By Theorem 10.1, it is enough to establish the finiteness of (2) for one decreasing weight with divergent integral and one fixed baseline. This replaces the all-index mass by a discounted series supported only at LCM records.

Remark. One further question is not left open here, since it is answered in [7]: whether normalised vanishing follows from $a_{n+1}/a_n^2 \rightarrow 1$ and rationality. It does. After deleting a finite prefix, Koizumi's Corollary 10 [7, p. 8] makes the sequence the pseudo-greedy expansion of its own reciprocal sum and gives $\varepsilon_n \rightarrow 0$ under exactly the hypotheses of Problem 1.1; rationality is not needed for that step, only summability of $\sum 1/a_n$. Since $\varepsilon_n = E_n/C_n$ under the intended reading of Section 2, this is normalised vanishing. Strict centring comes from the same source and is stronger than what Theorem 8.1 assumes:

for a rational reciprocal sum, the range $-c_n/2 \leq e_n < c_n/2$ of his Lemma 4(2) [7, p. 11] gives $|E_n| < C_n$ at every index, which is stronger than eventual strict centring. The eventual form also follows directly from normalised vanishing with $K = 1$. What remains unsupplied is hypothesis (5), the bound on the negative part, which is why the theorem is still conditional.

11.4 A barrier extension

Problem 11.6 (variable-rise CRT barrier). Determine the weakest growth condition on the positive increments of u_n under which an unbounded integer sequence cannot remain coprime to infinitely many pairwise-coprime old moduli as in Theorem 7.3. In particular, does the conclusion remain valid under the candidate condition

$$(u_{n+1} - u_n)_+ = o(\log \log(u_n + 3))?$$

The displayed rate is a proposed threshold, not a proved sharp boundary. Either an extension of the CRT barrier or a counterexample with this rise rate would be informative independently of Problem #243.

Formalisation targets

Erdős–Straus. Formalise Theorem 3 of [1], with its prefix-LCM quotient and correctly indexed growth factor, under the published analytic hypotheses.

Duverney. Formalise Corollary 3.2 of [2], including its signed numerators and its one-sided summability hypothesis, and then recover the all-positive specialisation used here. These are useful verification projects, but neither is an additional open mathematical problem.

Statements and declarations

Artefact and data availability. The [pinned formal-source revision](#) contains the Lean sources, the fixed toolchain, and the library manifest used in the verification. Proof authority remains with those sources; this manuscript provides navigation and exposition.

Lean checks each proof term against the fixed library version, and the sources linked here contain no proof placeholders and no project-defined axioms; Lean does not authorise the exposition, the citation choices, or the interpretation, for which the author remains responsible.

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A Guide to the formal sources

Each linked phrase opens its Lean declaration at the pinned source revision 99f4bf47422a. The state system, the exclusions, the barrier, and the bounded-negative-part theorem are in `ReciprocalTailRigidity.lean`; the finite-negative-mass

theorem is in `SparseResetRecovery.lean`; and the repair-entropy inequalities are in `RepairEntropy.lean`. The residue reduction of Appendix B is in `FiniteHorizon-Residue.lean`. Three distinctions are worth carrying into the source. The identification of any of these modules with reciprocal tails is the exposition of Section 2 and is not a checked statement. The periodic exclusion assumes the regime $e_n < a_n$. And the final Lean declaration lists strict centring and normalised vanishing as hypotheses; the former follows eventually from the latter with $K = 1$, while the formal source derives neither from the original analytic problem. For the external bridge supplying them, see Sections 1 and 8.

B A factorial residue reduction for forced orbits

In the case $m = c = 1$ of Section 5, where the shape equation (5.1) reads $D_n + 1 = (a_n - 1)(n + 1)$, each multiplier is determined by its predecessor; we call such an orbit *forced*. At index n the numerator of the next multiplier is

$$\text{num}(n, a) = (n + 1)a^2 - (n + 2)a + (n + 3),$$

the *forced numerator*, and the divisor is $n + 2$. The orbit survives a step when that division is exact, giving a survival predicate, the *survival predicate*. Example 5.1 is the forced orbit from $a = 3$ read this way: $\text{num}(0, 3) = 6$ is divisible by 2 and gives $a_1 = 3$, while $\text{num}(1, 3) = 13$ is not divisible by 3, so the orbit stops there. Deciding survival by iteration is expensive because the orbit grows doubly exponentially, and it is unnecessary: survival over a finite horizon depends on the initial value only through a factorial residue.

Theorem B.1 (factorial residue reduction). *For all h and all integers $a \equiv b \pmod{(h + 1)!}$, the orbit from a survives h forced updates exactly when the orbit from b does.*

Proof. Let $M(0, i) = 1$ and $M(h + 1, i) = (i + 2)M(h, i + 1)$, an ascending factorial with $M(h, 0) = (h + 1)!$. The numerator is a polynomial with integer coefficients, so congruences transfer; reducing the modulus gives divisibility by $i + 2$ for one exactly when for the other, and cancelling that common factor from both values and modulus leaves the inductive hypothesis at $M(h, i + 1)$. $\square$

Formalised as the *factorial residue reduction*, over the *shrinking-modulus transport*, the *polynomial congruence*, and the *exact-division cancellation*; the modulus identifications are the *ascending-factorial form* and the *factorial value at the initial index*.

At $h = 1$ the modulus is $2! = 2$, and surviving one update means $2 \mid a^2 - 2a + 3$, which holds exactly for odd a : for instance $\text{num}(0, 3) = 6$ but $\text{num}(0, 4) = 11$. So one step of survival is decided by the parity of a alone, which is Theorem B.1 at its smallest nontrivial horizon.

The search this supported is superseded. Running it over initial states below 5000 produced forced prefixes of length 17 and no longer; Theorem 5.2 now excludes the constant-negative case outright, for every seed and at every scale. The reduction is retained because it is exact, and because the shrinking-modulus technique transfers to any forced orbit whose step is a polynomial division.

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